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ORIGINAL ARTICLE

Cultural Exaptation and Cultural Neural Reuse: A Mechanism for the Emergence of Modern Culture and Behavior

Francesco d'Errico[1,2] · **Ivan Colagè**[3,4]

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Abstract

On the basis of recent advancements in both neuroscience and archaeology, we propose a plausible biocultural mechanism at the basis of cultural evolution. The proposed mechanism, which relies on the notions of *cultural exaptation* and *cultural neural reuse*, may account for the asynchronous, discontinuous, and patchy emergence of innovations around the globe. *Cultural exaptation* refers to the reuse of previously devised cultural features for new purposes. *Cultural neural reuse* refers to cases in which exposure to cultural practices induces the formation, activation, and stabilization of new functional and/or structural brain networks during the individual lifespan. The invention of writing is interpreted as a case of cultural exaptation of previous devices to record information, in use since at least the Early Later Stone Age and the beginning of the Upper Paleolithic (44,000 years before present). The measurable changes in brain structure and functioning caused by learning to read are proposed as an exemplar case of cultural neural reuse. It is argued that repeated cycles of cultural exaptation, development of appropriate strategies of cultural transmission, and ensuing cultural neural reuse represent the fundamental mechanism that has regulated the cultural evolution of our lineage. A general predictive model of when and under which circumstances the proposed mechanism should be expected to occur is proposed, and the relationship of our mechanism with gene-culture coevolutionary models is discussed.

Keywords Artificial memory systems · Brain plasticity · Cultural innovations · Paleolithic · Reading · Social learning

Introduction

A main challenge facing researchers who seek to understand how and when our ancestors developed cultures similar to those known nowadays and historically is that of identifying mechanisms to account for cultural evolution. Natural selection has certainly acted as a powerful factor shaping the evolution of life on our planet. The evolution of hominins has not escaped such governing rules. This is particularly true for members of our lineage due to their successive expansions in new latitudes and biogeographic regions, which has considerably expanded their ecological niches. It is widely accepted, however, that those expansions or successful local adaptations to climate changes would have been unlikely without the emergence of highly adaptive cultural innovations. Culture can be defined as a body of information acquired from other members of the population via teaching, imitation, or other forms of social interaction, which is capable of affecting the behavior of individuals (Richerson and Boyd 2005). Cultural *innovation* may be defined as any change in the cultural repertoire of a population resulting from a novel combination of existing cultural traits, new use of old items, or introduction of significant modifications in existing tools, techniques, or practices (Legare and Neilsen 2015). Cultural innovations may emerge by accident or error, and as a consequence of highly developed, efficient and flexible transmission strategies and teaching (Burdett et al. 2017). In general, a cultural innovation allows for the solution of a new problem, the exploitation of a new resource

✉ Francesco d'Errico
francesco.derrico@u-bordeaux.fr

1 UMR-CNRS PACEA de la Préhistoire à l'Actuel: Culture, Environnement et Anthropologie, Université de Bordeaux, Bordeaux, France

2 SFF Centre for Early Sapiens Behaviour (SapienCE), University of Bergen, Bergen, Norway

3 Faculty of Philosophy, Pontifical Antonianum University, Rome, Italy

4 DISF Centre, Pontifical University of the Holy Cross, Rome, Italy

(or a novel use of a known resource), or the implementation of a new behavioral repertoire.

It has been argued that cumulative culture (e.g., Tomasello 2001; Tennie et al. 2009; Dean et al. 2012) has allowed humans to greatly lessen, if not avoid, the impact of natural selection when adapting to the environmental changes that have characterized the last 500,000 years of the planet's climate (Richerson and Boyd 2005). Although gene-culture coevolution and the growing role acknowledged for it in our ancestors' history have become the reference framework for all those interested in understanding cultural change in an evolutionary perspective, this framework does not indicate, in and of itself, the mechanism that has allowed past human populations to reach the cognitive and cultural complexity that we observe nowadays. If genetic evolution alone were responsible for the emergence of cumulative culture this would entail a process of isolation, production of random mutations, and selection of successful variants leading to novel cognition, adapted to typically human transmission and accumulation of knowledge. This scenario was long advocated for by those who considered modern cognition and rapidly ensuing modern culture as the outcome of a switch in neural constitution due to a genetic mutation occurring 50 kya (Mellars and Stringer 1989; Klein 1989, 2000; Tattersall 1995; Bar-Yosef 1998). The modern character of the Upper Paleolithic cultural traditions closely following that mutation would have left little space for gene-culture interaction to shape that complexity. This scenario was challenged by a string of archaeological discoveries at Middle Stone Age (MSA) sites from northern and southern Africa revealing that a number of cultural innovations possibly reflecting modern cognition (complex hafting techniques, formal bone tools, refined stone knapping and tool retouching, personal ornaments covered with ochre, abstract engravings, systematic production of pigment) occurred in that continent well before 50 kya (McBrearty and Brooks 2000; Barham and Mitchell 2008; d'Errico and Stringer 2011; d'Errico and Banks 2013; Scerri 2017). This evidence has been generally interpreted as supporting a scenario according to which the selective process that has given raise to our species in Africa would have granted this species new cognitive functions favoring the gradual and progressive emergence and accumulation of innovations, including the production of a symbolic material culture. Contrary to the previous stochastic scenario for the emergence of cumulative culture, this scenario would give a more prominent role to the interplay of genes and culture, as most of the cultural innovations observed in the African MSA occurred dozens if not hundreds of thousand years after selection of the adaptive mutations producing our species.

If, however, mechanisms of gene-culture coevolution similar to those observed in extant human populations came into existence only after the origin of our species in Africa, we should not find the suite of innovations characteristic of the African Middle Stone Age outside that continent before the extensive spread of modern humans out of Africa ca. 70–60 kya. A growing body of archaeological discoveries and reappraisal of old finds instead suggest that many key cultural innovations were developed outside the African continent before 70–60 kya (see Colagè and d'Errico 2018 for an overview). Established ability to ignite and control fire, complex and changing lithic technologies, elaborated wooden weapons, demanding hafting techniques, varied hunting strategies enabling archaic hominins to kill dangerous game, exploitation of marine and plants resources, organization of living space, and seafaring are all innovations now attributed to Neandertal cultures in various regions of Europe, before any contact with modern humans (d'Errico and Stringer 2011; Johansson 2014; Villa and Roebroek 2014). Other lines of evidence—burials or mortuary practices more generally, collection of rare items, pigment use, engravings and paintings on objects and cave walls, and the extraction of bird feathers and claws—support the notion that Neandertals were engaged in symbolically mediated behavior, well before any contact with anatomically modern humans (Pettitt 2011; Rodríguez-Vidal et al. 2014; Villa and Roebroeks 2014; Zilhão 2016; Majkić et al. 2017; Stiner 2017; Hoffmann et al. 2018). The production of formal bone tools, sometimes decorated with notches and a varied array of personal ornaments, found at late Neandertal sites, demonstrates that the production of items characteristic of the Upper Paleolithic was fully in the realm of Neandertal cognition (d'Errico et al. 2003; Soressi et al. 2013). In addition, isolated, puzzling evidence for mortuary practices (Pettitt 2011; Dirks et al. 2015, 2017; Zilhao 2016) and abstract engravings (Mania and Mania 1988; Joordens et al. 2015) occur in and outside Europe at sites that are several hundred thousand years older than the Mousterian cultures of Europe and the Near East. Paleogenetic data demonstrating interbreeding between Neandertals, modern humans, and Denisovans (Fu et al. 2016), and suggesting gene flow from an archaic African population into the modern human genome (Hammer et al. 2011), show that possible differences in cognition produced by genetic drift and natural selection apparently did not prevent successful social integration and reproduction of hybrids.

The above evidence is inconsistent with a scenario according to which cumulative culture is the outcome of a new cognitive architecture stemming from a single speciation event occurring in Africa 260 kya (Schlebusch et al. 2017). Rather, it suggests that genes and culture coevolved for much longer than may have been expected, and that they interacted in ways still unknown to us.

Recent work has enquired into this issue, and particularly into the correlations between the appearance of cultural innovations and a number of social/demographic or ecological/climatic factors (Shennan 2001; Henrich 2004; Powell et al. 2009; Premo and Hublin 2009; Kolodny et al. 2015; Collard et al. 2016).

By applying modeling techniques, these authors show that population size, density, rates of reproduction, networks of information exchange, and climate change can determine whether or not cultural innovations can be diffused, maintained, and in some cases lost. The interest of their results is that their predictions better fit the archaeological record than models proposing abrupt or incremental changes in behavior. However, a possible limitation of these proposals is that they imply that factors such as demography and climate were only effective in populations to which evolution granted a novel cognition, i.e., anatomically modern humans. Under this view, genetic evolution (a speciation event) remains the prime mover. The above proposals are complementary to the one we will develop in the following, in that they address the "extrinsic" factors (demography, climate, etc.) for cultural innovations to emerge, while we will focus on the intrinsic factors (exploiting old solutions for new problems, strategies for knowledge transmission, plastic potential of the brain) that allow the former to be effective.

Investigating these intrinsic factors is necessary since it is still unclear how the key cultural innovations that make us human and had profound effects on our ancestors' lifestyles may have emerged and been transmitted, or sometimes lost, without this being caused by—or directly linked to—heritable genetic changes or speciation events. A unified theory accounting for the evidence summarized above is needed as it would unveil the mechanisms responsible for the emergence, maintenance, and loss of cultural innovations—or, in a nutshell, for how we became humans.

An interesting perspective is proposed by Heyes (2012, 2017, 2018), who finds little evidence that genetic adaptation is at the basis of the mechanisms that have allowed cumulative culture. She argues in favor of the idea that most cognitive processes enabling cumulative cultural evolution—social learning, imitation, teaching, social motivation, theory of mind, and, in particular, reading—are essentially cultural in nature and did not stem from genetic evolution.

The goal of this article is to take a further step towards a unified theory of cultural evolution by proposing two plausible *mechanisms* lying beneath the emergence of cultural innovations: "cultural exaptation" and "cultural neural reuse." *Cultural exaptation* refers to the reuse of previously devised cultural features for new purposes. *Cultural neural reuse* refers to cases in which exposure to cultural practices induces the formation, activation, and/or stabilization of new functional and/or structural brain networks during the individual lifespan. Our specific focus is on the mechanisms that make it possible for a cultural innovation to emerge within a society, stabilize, spread within and beyond a group, and become detectable in the archaeological record.

Cultural Exaptation

The notion of exaptation was introduced by Gould and Vrba (1982) in the field of evolutionary biology to indicate traits that have not been selected for their current function, but that evolved either for another function or for no specific function. The key point of biological exaptation is that, in order to implement a new biological function, waiting for genetic evolution to provide a dedicated structure is not always needed. A preexisting structure may be co-opted for the new function even when the previous evolution of the co-opted structure is unrelated to the novel function. After a preexisting trait is exapted for a new function, that trait evolves also according to the constraints imposed by the new function, and is progressively refined by genetic variation and natural selection.

A classical example of exaptation is that of birds' feathers (Gould and Vrba 1982), which seemingly first emerged for thermoregulation and were only later "exapted" for flying, and successively refined and diversified for the various flying needs of their carriers. Another example may be junk DNA, which may be initially accumulated without any functional import and may later on acquire a functional role in gene-expression regulation, and eventually be submitted to ensuing selective pressures. In general, the old function of an exapted trait does not need to disappear, as the exapted trait may keep performing the old function as well.

Cultural exaptation applies this idea to cultural innovations and refers to the co-option of existing cultural features for new purposes. Cultural exaptation, as understood here, does *not* require concomitant biological exaptation or heritable genetic changes as a prerequisite, but occurs at the truly cultural level. In analogy with biological exaptation, however, the co-opted cultural feature may have been originally invented for purposes quite different from the one for which it later became "culturally exapted." In some cases, as we will see shortly, a utilitarian cultural feature may be co-opted for nonutilitarian, aesthetic, or symbolic aim.

The use of ochre, natural earth pigment containing hydrated iron oxides, may provide an example. It has been shown that compounds made with ochre are powerful photoprotective means to shield the skin from harmful ultraviolet radiation (Rifkin et al. 2015). There is evidence suggesting symbolic use of iron-rich pigments on personal ornaments since at least 80 kya in Africa (d'Errico et al. 2009; d'Errico and Backwell 2016). One can envision a scenario in which ochre was first exploited for its photoprotective function and, in a later phase, through cultural exaptation, became a means, for example through body painting, to reinforce cultural mechanisms related to intra- and inter-group self-identification leading to the emergence of diversified symbolic material cultures. A subsequent cultural exaptation may have occurred in an already fully symbolic context, when ochre was purposely employed on ornaments and clothes. As with biological exaptation, the old function of the culturally exapted trait does not necessarily disappear after exaptation.

Cultural Neural Reuse

The notion of neural reuse has recently been introduced (Anderson 2010, 2014) to indicate that for a new cognitive function to emerge it is not always necessary to evolve *new* specific brain areas. A preexisting brain area or region—already evolved for a previous function—may be reused and put at the service of a new cognitive function, provided that the basic information-processing capabilities (or "workings") of the reused region are apt to fulfil the new function. Neural reuse is a specific form of *biological* exaptation concerning brain areas. As in "ordinary" biological exaptation, the reused brain area then undergoes evolutionary refinement constrained by the new role it assumed (Anderson 2007; Colagè and D'Ambrosio 2014). Once reused, the brain area likely begins to work in a novel (or at least partially different) brain network with respect to the one where it worked previously, so that not only the reused area, but also the newly resulting network begins to evolve, via the ordinary mechanism of genetic variation and natural selection, under the constraints posed by the new function and the specifics of the network.

A good example of neural reuse is the Broca's region. This region: (1) is involved in several different functions (action execution, observation and imitation, imagery tasks, music perception, and language), (2) emerged at different evolutionary times (Anderson 2010), and (3) shows a complex subdivision (Amunts et al. 2010) in several cyto- and receptor-architectonically distinct areas. Moreover, the white-matter fiber tracts connecting it with other distant areas underwent substantial evolution from macaques to chimpanzees to humans (Rilling et al. 2008).

C*ultural* neural reuse differs from the general, or "non-cultural," neural reuse in that it refers to a reuse of brain areas that happens at the ontogenetic timescale (Dehaene and Cohen 2007; Colagè 2013, 2015; Colagè and D'Ambrosio 2014). Cultural neural reuse alludes to cases in which exposure to cultural practices induces the formation, activation, or stabilization of new functional and/or structural brain networks during the individual lifespan.

We identify four conditions for something to be qualified as an instance of cultural neural reuse:

(a) A brain area or region is to be reused for a cognitive/functional domain (e.g., language, reading, numeracy, etc.) distinct from the one(s) it initially subserved (e.g., action recognition, object recognition, etc.), and for which it evolved.
(b) The reuse has to result in the formation or stabilization of a *new* (functional or anatomical) network in the brain.
(c) The reuse must not be the direct outcome of heritable modifications in the genetic-developmental program that controls the formation of the species-specific brain gross functional anatomy and structural connectivity.
(d) The reuse is to be induced or triggered by the exposure to pervading cultural practices, involving teaching or other means of cultural transmission.

Cultural neural reuse refers to specific instances of *neuroplasticity*. Neuroplasticity indicates the ability of the brain, and the central nervous system more generally, to modify physiological, functional, and/or structural features as a consequence of experience and practice, or following brain lesions. In this way, the brain is able to adapt plastically to new requirements coming from new sensory inputs, new behavioral outputs, new information-processing needs, or in the face of brain damage, virtually all along the individual lifespan, though to different degrees in different periods (Pascual-Leone et al. 2005).

Neuroplasticity manifests itself at different levels. The most basic one is the local cortical level: learning new skills brings about structural changes in the volume of gray matter. For example, the volume of posterior hippocampal gray matter increases in taxi drivers that have learned the layout of London's 25,000 streets (Woollett and Maguire 2011). Such gray-matter increases might be putatively interpreted as an effect of synaptic strengthening. Now, although the taxi driver example has striking features, it is an instance of something happening regularly when new specific skills are learned, as in virtually all cases of perceptual or motor learning. However, what happens in such cases is the refinement and/or potentiation of *already existing* circuits (like spatial memory circuits in the taxi drivers' example). Cultural neural reuse, on the contrary, intends to refer to (and define) a subclass of cases of neuroplasticity in which *new brain networks* are formed following the acquisition of a new cultural ability.

A brain network, quite generally, is a set of several distinct brain areas, often scattered across the cortex, altogether involved in a certain function. For example, though language is thought to involve more and more portions of the cortex, the so-called "core language network" is usually considered as composed of superior temporal (Wernicke's), inferior parietal (Geschwind's), and inferior frontal (Broca's) regions. There are two main ways to define a brain network: (1) a set of brain regions connected by white-matter fiber pathways (anatomical or structural connectivity), which establish an *anatomical* network (Sporns et al. 2005; Hagmann et al. 2008); (2) a set of brain regions showing specific patterns of coactivation (functional connectivity), thus reflecting a *functional* network (Horwitz et al. 1999, 2000). These two views are not contradictory, and cultural neural reuse, as we define it here, may result in the formation of a new functional and/or anatomical network.

Therefore, cultural neural reuse refers to cases in which new (functional or anatomical) networks are formed during the lifespan of an individual who acquires a specific cultural skill through teaching or dedicated training. These changes are not heritable.

From Artificial Memory Systems to Writing: Cultural Exaptations

We will now focus on the cultural innovation consisting in the invention of systems of notations, i.e., devices that often take the form of incised objects used for storing, recovering, and possibly transmitting information recorded outside the body: hence the label of "artificial memory systems" (AMSs). Starting in the 1960s, proposals have been made to consider a number of Upper Paleolithic and Mesolithic objects as AMSs (Marshack 1964, 1972a, b). In the following decades, given the impossibility of understanding what information was actually stored on AMSs on the basis of the archaeological record and in the absence of ethnographic data, the focus became that of creating a theoretical framework to identify the *type* of code used by an AMS instead of the actual information recorded. A survey of AMSs presently in use in different communities worldwide and distinct from writing systems individuates four distinct *factors* for information coding (d'Errico 1995, 1998, 2002): (1) the number of marks on the object, (2) the accumulation of marks on the object over extended periods of time, (3) the spatial distribution and arrangement of marks on the object, and (4) the morphology (or the different morphologies) of the marks. It was also shown that establishing whether an object is to be considered an AMS requires a technological analysis of the marks. Sequential markings on Paleolithic objects are made by engraving single or multiple stroke lines with sharp points, cutting notches by a unique or to-and-fro movement of a cutting edge, and puncturing a surface with a robust point by pressure, percussion, or rotation. The microscopic analysis of marks produced experimentally with these techniques has identified criteria to establish the technique used, the order in which the marks were made, and to distinguish, for example, sets of marks produced by the same tool in a single session from sets representing an accumulation of marks made with different tools, probably at different times. It has also allowed us to identify whether specific techniques have been applied to produce marks of different morphology, possibly bearing different meanings.

D'Errico (2002) provides detailed clues to establish in which cases the application of these experimental criteria allows for a firm identification of prehistoric AMSs, those in which this interpretation is plausible but cannot be considered as definite, and those that are better interpreted as decorations, marks of ownership, and the like. For example, in incised objects used as AMSs with codes based on the accumulation of information over time, changes of tool, repeated inversion in the position of the marking tool, and changes in the marks' spacing and orientation will occur between each new stage of marking, i.e., every time the information is updated by adding new marks. These objects will also bear diagnostic use-wear due to long term curation and handling (d'Errico 1993).

Application of these criteria to Paleolithic objects bearing sequential markings has shown that a number of them can reasonably be interpreted as AMSs (d'Errico 1998). The earliest known artifact for which such an interpretation was tentatively proposed (d'Errico et al. 2017) is a mesial fragment of a hyena femur found at the Neandertal site of Les Pradelles, France, in layers dated ca. 72–60 kya. It bears nine parallel deep incisions and two groups of small, superficial, partially overlapping incisions located at the bottom of the second and third main incisions. Microscopic analysis of the main incisions revealed that they were made with the same tool in rapid succession. Experimental reproduction of this pattern by naive subjects has shown that the Neandertal craftsman intended to produce similar incisions, without aiming at producing a visual pattern based on equidistant marks. This evidence fits the interpretation of this set of incisions as recording numerical information enriched with diacritic signs.

More recent is a broken baboon fibula (Fig. 1a), marked with 29 notches cut with five different tools. Found at Border Cave, South Africa, in Early Later Stone Age archaeological levels dated to 44–42 kya, this object is interpreted as an AMS with a code based on the accumulation of marks over time (d'Errico et al. 2012, 2017). The same interpretation is proposed for a reindeer metapodial (Fig. 1b) from the Gravettian layers of the Labattut shelter, Dordogne, France, bearing 65 notches cut with three or four different tools (d'Errico 1998). A rib of a woolly rhinoceros from Solutré, Saône-et-Loire, France, with 53 notches produced by at least twelve different cutting edges (d'Errico 1998) is also interpreted as an AMS with a code based on the accumulation of marks over time (Fig. 1c, d). The information recorded on these three objects can be recovered by visual and/or tactile inspection. A rib from the Magdalenian layers of the Laugerie-Basse shelter, Dordogne, France, is incised with more than 120 marks grouped in at least ten different sets (Fig. 1e). The code is possibly based on accumulation of marks over time, complex spatial organization of the marks, and marks morphology. The number and size of the marks suggest that the information stored on this

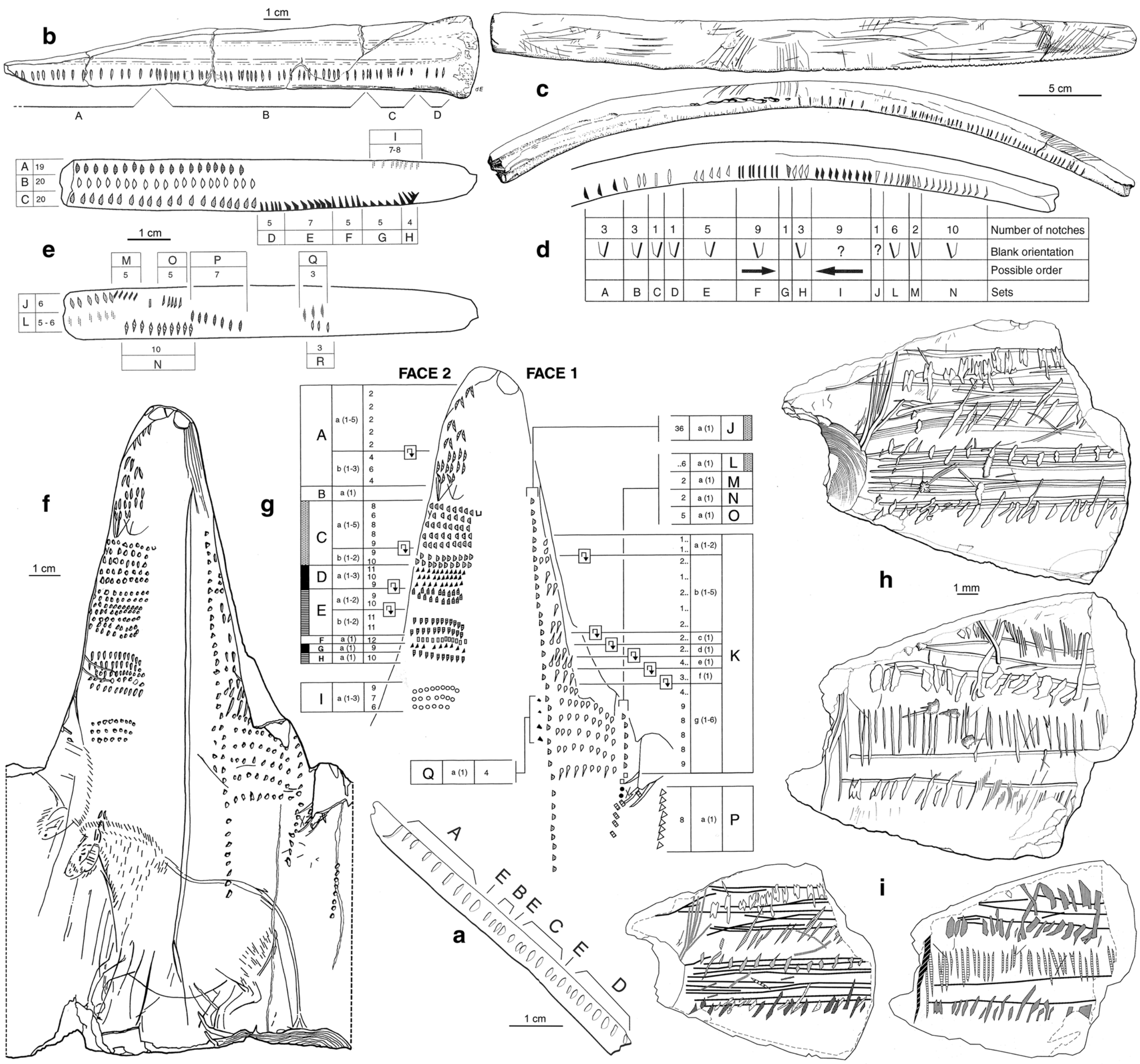

Fig. 1 Incised Paleolithic objects interpreted as artificial memory systems; **a** baboon fibula from Border Cave, South Africa; **b** reindeer metapodial from the Labattut rock shelter, Dordogne, France; **c, d** rib of a woolly rhinoceros from Solutré, Saône-et-Loire, France, and its schematic rendering; **e** rib from Laugerie-Basse shelter, Dordogne, France; **f, g** reindeer antler from La Marche, Vienne, France, and its schematic rendering; **h, i** fragment of a bone pendant from Tossal de la Roca, Alicante, Spain, and its schematic rendering. Capital letters and patterns identify sets of marks produced by different tools and/or techniques. (Modified after d'Errico 1995, 1998; d'Errico et al. 2012; d'Errico and Cacho 1994)

object was retrieved visually. The rib also displays signs of deliberate obliteration of three sets of marks, a behavior incompatible with a decorative purpose. A reindeer antler (Fig. 1f, g) from La Marche, Vienne, France, bears more than three hundred marks arranged in several rows and spatially distinguished sets (d'Errico 1995). Each set of marks is produced with a slightly different technique in order to create small but visually perceptible differences between sets. Some distinct diacritic marks are incised at the beginning of some sets, probably to provide additional cues to grasp the meaning of the marks composing the set. The object is interpreted as an AMS with a code based on both the spatial distribution and the morphology of the marks. The small size of the marks and the presence of faint traces of red pigment, added to enhance morphological differences between marks, suggest that information was retrieved visually. A fragment of a bone pendant (Fig. 1h, i) from the late Magdalenian site of Tossal de la Roca, Alicante, Spain, is engraved with eight sets of tiny marks, each set made with a different technique or tool (d'Errico and Cacho 1994). This object can be interpreted as an AMS based on the spatial distribution of elements, element morphology, and possibly accumulation of marks over time. Given the high number of marks on a very small surface, the retrieval of information could have been done only *visually*.

Only careful technological analysis of objects' marks relying on clearly defined criteria may help distinguish AMSs from pieces incised for other reasons. This is due to the fact that non-AMS incised pieces share a number of characters with AMSs: the materials, tools, and engraving techniques used for AMSs are the same used for many other Paleolithic objects with utilitarian or decorative purposes. This suggests that the emergence of AMSs may be interpreted as a case of cultural exaptation in which materials, tools, and techniques developed for producing objects without any notational import have been co-opted for a new purpose, namely, the production of AMSs. At a more "cognitive" level, Upper Paleolithic AMSs emerge when personal ornaments, abstract engravings, and object coloring were already in place. This may suggest that also the cognitive ability to relate artificial objects or objects' features to some kind of "meaning" (self-identification, social status or belonging, ownership, etc.) was already developed and then exapted for producing AMSs. The limited number of AMSs identified in the archeological record may indicate that these objects were only used by selected members of the Upper Paleolithic hunter-gatherer communities. This possibility does not dismiss the idea that AMSs represent a case of cultural exaptation, since their invention represented a novelty having significant consequences on the sociocultural setting of those communities.

Many AMSs use morphological differences between marks to make the recording and retrieval of information possible. The marks, therefore, had to be recognized and associated with their meaning. Though we do not know (and probably will never know) the meaning of the marks on Paleolithic AMSs, it is reasonable to assume that visually homologous marks were indicating the same type of item, and that morphological differences were used in some codes (mainly at the end of the Upper Paleolithic) to indicate different items. As will be argued below, the visual recognition and discrimination of marks' morphology, and the association of marks with specific items (and, consequently, with some "meaning") closely resemble processes implied in reading proper writing systems. This will prompt further considerations on the relationship between cultural exaptation and cultural neural reuse, and between Upper Paleolithic AMSs and proper writing systems.

Reading and Cultural Neural Reuse

The invention of writing may be considered as the most recent and fundamental turn in the human way of living and thinking (Goody 1977; Donald 1991; Ong 2002). In the last two decades, several neuroscientific studies have inquired into the neural substrates of the human ability to read. Reviewing some key aspects of the neuroscience of reading will unveil two main points. First, the invention of writing may also depend on the development of more ancient AMSs, i.e., writing might have emerged as a cultural exaptation from previous AMSs. Second, learning to read brings about several measurable changes in brain structure and function that cannot be considered as the direct outcome of proximate heritable genetic changes, but are consistent with cultural neural reuse as defined above.

The so-called dual-route model (Warrington and Shallice 1980; Coltheart and Rastle 1994) is seen as the reference model of reading. It distinguishes two neural routes employed to retrieve meaning from written language. The first one, called the "direct" or "lexico-semantic" route, retrieves the meaning of a word without passing through its phonology (pronunciation) but just by recognizing the word's visual form as a whole and linking it directly with semantic information. Besides the obvious activation of early visual occipital areas, the direct route comprises regions in the left occipito-temporal junction, as well as areas subserving semantic access: left basal temporal area, left posterior middle temporal gyrus, and the triangular part of the left inferior frontal gyrus (Jobard et al. 2003). The second route is called "indirect" or "grapho-phonological." It is mainly engaged in situations when words' visual form cannot be readily recognized, e.g., highly infrequent or very long words or words never previously encountered. In such cases, the reader has to translate the orthographic stimuli into phonological ones in order to recognize the word and retrieve its meaning. At the neural level, apart from early visual areas, this route engages the left occipito–temporal junction (like the direct route), as well as a number of left-lateralized areas implicated in phonological processing: superior and middle temporal gyri, posterior superior temporal gyrus, supramarginal gyrus, and the pars opercularis of the inferior frontal gyrus (Jobard et al. 2003). After phonological processing, semantic access is ensured by the same areas as for the direct route.

Skilled and efficient reading requires the proper functioning of both routes. Specifically, the direct route ensures the rapid retrieval of semantic information from orthography, at least for well-known words and/or in languages in which the connection between orthography and phonology does not follow clear and stable rules—e.g., English versus Italian. The indirect route is crucial for infrequent words whose meaning must be retrieved through phonology. It plays a key role also when learning *any* written word for the first time. Indeed, phonology is essential in the first stages of acquisition of literacy also in nonalphabetic languages, such as Chinese (Tan et al. 2005).

The main difference between Upper Paleolithic AMSs and fully developed writing systems likely lies in their link with *phonology*: ancient AMSs do not show features suggesting that they are graphical renderings of spoken language, whereas any historically known writing system constitutes a graphic instantiation of a spoken language. Moreover, we have seen that the information stored in prehistoric AMSs was likely retrieved via visual or tactile inspection of marks' morphology and arrangement. This might suggest that the AMSs mainly involved the direct lexico-semantic route well before the need for the indirect one required by proper writing systems, and that, via cultural exaptation, the invention of writing exploited the ability to link visual marks to semantic information and joined this ability with that of linking visual marks with phonological information (Fig. 2).

It is worth noting that both routes comprise the left occipito–temporal junction, known to host the so-called visual word form area, or VWFA (Jobard et al. 2003; Cohen et al.

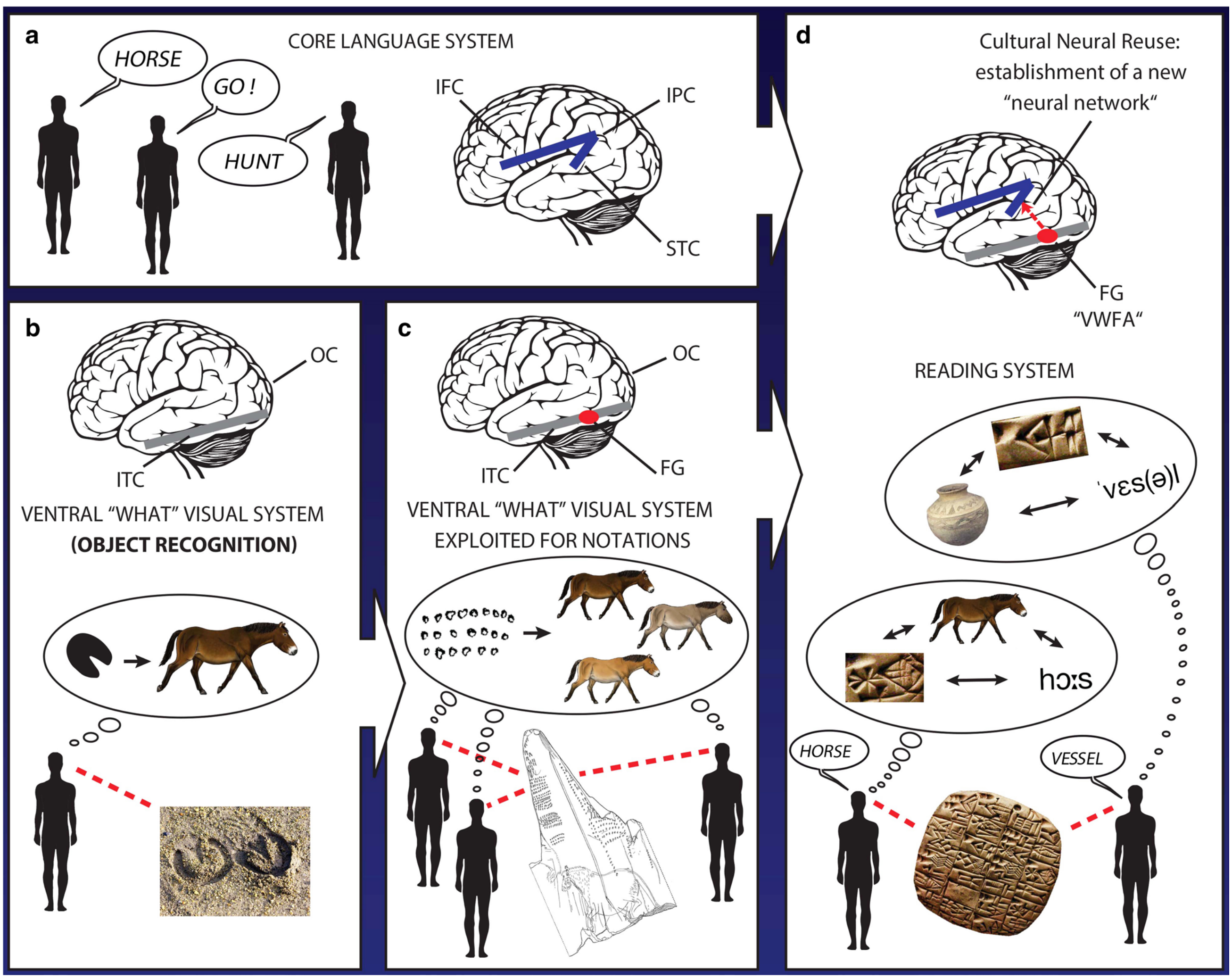

Fig. 2 Representation of the possible process leading to the invention of writing systems involving cultural exaptation and cultural neural reuse. Language and the related brain network (**a**) were already in place much earlier than the invention of writing systems. The ventral visual path for object recognition (**b**) certainly predates the appearance of the genus *Homo*. During the Early Later Stone Age and the Upper Paleolithic, the ventral visual pathway was likely exploited to give rise to prehistoric systems of notation (**c**). The invention of writing implies the cultural exaptation of both spoken language lexicon and systems of notation (**d**, bottom). At the neural level, according to cultural neural reuse, reading requires the ontogenetic formation of a new network integrating the ventral visual pathway for object recognition and the language system (**d**, top). The VWFA constitutes the key brain region for such an integration. *FG* fusiform gyrus, *IFC* Inferior frontal cortex, *IPC* inferior parietal cortex, *ITC* inferior temporal cortex, *OC* occipital cortex, *STC* superior temporal cortex, *VWFA* visual word form area

2000). This brain region is nowadays considered a key reading area.

The first decisive evidence about the role of this area came from a study (Cohen et al. 2000) on patients with lesions in the posterior *corpus collosum* (the white-matter fiber bundle connecting the two brain hemispheres). When orthographic stimuli were presented in the left visual hemifield (projecting to the right visual cortex, contralateral to the VWFA), the controls had normal reading performance and fMRI activation, whereas patients' reading performances dropped dramatically and no activation in the VWFA was recorded. The callosal lesion prevented the visual information from reaching the (left) VWFA from the right occipital cortex, thus hindering reading and showing the causal role of the VWFA for such a skill.

Moreover, a study with an epileptic patient (Gaillard et al. 2006) further suggested the essential causal role of the VWFA in reading. As tested prior to surgery, the patient's reading skills were normal, and the VWFA activated in response to visual words. The patient then underwent surgery removing a very circumscribed brain region (the source of epilepsy) whose anterior edge was just posterior to the VWFA peak-activation site. After surgery, the patient developed a severe form of pure alexia, and the VWFA activation in response to words no longer appeared.

Though the specific role of the VWFA in reading is debated (Price and Devlin 2003; Vigneau et al. 2005), there is increasing consensus that the VWFA in literate individuals is specifically dedicated to identifying orthographic stimuli and associating them with phonological and lexical information (Dehaene et al. 2015), in keeping with its involvement in both the lexico-semantic and grapho-phonological routes discussed above. Studies (Baker et al. 2007; Szwed et al. 2014) have also shown that the VWFA responds selectively to orthographic stimuli in the script the readers have learned (English, Hebrew, Chinese, etc.). Moreover, it has been shown that the VWFA's response to scripts is enhanced not just by visual exposure to orthographic stimuli, but specifically when the focus is on orthography-to-phonology correspondence, both in children and adults (Hashimoto and Sakai 2004; Brem et al. 2010).

It has been proposed (Dehaene and Cohen 2007) that the specification of the VWFA and its attunement to orthographic stimuli is the outcome of a process of acculturation, i.e., exposure to intensive and prolonged training to read and write. This idea is intended to harmonize two apparently conflicting points (Dehaene and Cohen 2011): (1) the strong specificity of the VWFA for reading, and (2) the fact that the specification of the VWFA cannot be the outcome of concomitant heritable genetic changes (i.e., that we cannot have "evolved for reading"). Indeed, reading started less than 6000 years ago (with the Mesopotamian cuneiform) and just one century ago only a tiny minority of human beings were literate: this contradicts the idea that the emergence of literacy required proximate heritable genetic changes inducing species-specific neural substrates for reading. It has nevertheless been, since its invention, a key cultural exaptation having broad consequences on the sociocultural setting of past peoples and inducing cultural neural reuse in those who learn to read.

Reading is therefore a perfect case of *cultural neural reuse* as defined in our Introduction. It is not the outcome of genetic evolution but mainly the consequence of ontogenetic exposure to an articulate cultural strategy transmitted through teaching—see *points* (*c*) *and* (*d*) of our definition. This is also supported by the fact that the specification of the VWFA happens with virtually the same features also when reading is acquired in adulthood (Dehaene et al. 2010). The VWFA as an area for reading exploits a very specific brain region (the VWFA) evolved for other functional purposes such as object and face recognition—*point* (*a*) of our definition. Recently, indeed, a subregion of the posterior fusiform gyrus has been cyto-architectonically defined, and labeled "area FG2" (Caspers et al. 2013). Interestingly, area FG2 has been shown to overlap largely with the VWFA in the *left* hemisphere of adult literate individuals and with the fusiform face area (FFA) in the right hemisphere (Caspers et al. 2014). Moreover, whereas the right FG2 (overlapping the FFA) mainly coactivates with the amygdala (involved in processing face emotional expression), the left FG2 (coextensive with the VWFA) mainly coactivates with other core language areas (superior temporal, inferior parietal, and inferior frontal) in the left hemisphere (Caspers et al. 2014). This might suggest that the FG2 (both on the right and left) likely evolved to be the FFA (as recognition of face emotional expressions is crucial in primates' lifestyle), but it becomes attuned to orthography in the left hemisphere (becoming the VWFA) only following the *ontogenetic* acquisition of reading skills (Cantlon et al. 2011).

Finally, there is evidence suggesting that learning to read affects the brain not only at the local level of the VWFA, but also at the level of network and interregional connectivity—*point* (*b*) of our definition. To begin with, literacy enhances the interhemispheric structural connections linking the left and right angular gyri through the corpus callosum (Carreiras et al. 2009). Moreover, the fractional anisotropy of the left arcuate fasciculus and the inferior longitudinal fasciculus (two intrahemispheric white-matter fiber bundles connecting temporal, inferior parietal, and inferior frontal areas) correlates with the reading proficiency of 7- to 15-year-old children (Yeatman et al. 2012). As fractional anisotropy is a measure of the efficiency of the communication among distant brain regions through white-matter, long-range axonal connections, it suggests that learning to read enhances the connectivity among visual, auditory, and language-specific brain areas. The fractional anisotropy of the temporo–parietal portion of the left arcuate fasciculus has also been shown to correlate with reading proficiency both in individuals that acquired reading at school age, and in individuals who learned to read only in late adulthood (Klingberg et al. 2000; Thiebaut de Schotten et al. 2012). Therefore, learning to read results in the formation, potentiation, and/or stabilization of a brain network that becomes specifically responsive to reading. Consequently, the specification of the VWFA, and its attunement to orthographic stimuli, likely depends on the process embedding this region in a brain network that forms in the human brain as individuals learn to read (Fig. 2). This view is consistent with an extended conception of epigenesis and, more generally, of phenotypic plasticity (D'Ambrosio and Colagè 2017).

Discussion

The cultural innovations that we recognize as being the cornerstone of typically human behavior emerged at different times and places over at least the last 500,000 years, among morphologically different human populations. This pattern challenges both the idea that a unique "genetic revolution" prompted the emergence of modern culture, and the view that the accretion of innovations observed in the archaeological record is directly due to single or multiple speciation events.

The complex pattern of emergence of cultural innovations prompts the notion of *cultural exaptation*: a novel cultural practice emerges by exploiting existing skills and techniques in new ways. The changing functions of mineral iron oxides (such as ochre pigments), discussed in our Introduction, is one example of the links that can be identified between succeeding cultural innovations. The improvement of pyrotechnology in the Southern Africa MSA, for example, has arguably allowed for the refinement of knapping techniques, hafting compounds, and the heat treatment of personal ornaments. The increased use of ivory and antler in the early Upper Paleolithic of Europe (ca. 42 kya) led to the production of diversified tool types, barbed points for fishing, needles, spear throwers, and a variety of personal ornaments. The refinement of techniques specifically conceived to modify mammalian hard tissues (sawing, scraping, grooving, incising, and engraving) at the beginning of the Upper Paleolithic has given rise to hafting and tool decoration techniques instrumental to conceiving AMSs.

AMSs exploited two sets of previously developed practices. The first comprises an array of tools and techniques to incise objects of the same materials used to make AMSs. The second involves the cognitive ability to associate material objects (like tools, personal ornaments, etc.) with some meaning. The joint exploitation of these two sets of skills made possible the conception and production of AMSs.

We have also reviewed key data and models in the neuroscience of reading. This showed that fluent reading requires the integration of two distinguishable brain systems: a lexico-semantic route linking the visual word form with meaning, and a grapho-phonological route converting orthography into phonology. The reviewed studies also suggest that reading acquisition deeply affects the brain, arguably promoting the formation of a dedicated network for reading, of which the so-called VWFA is an essential hub. Because the invention of writing is too recent to be an outcome of genetic evolution, we have regarded reading acquisition as a key instance of *cultural neural reuse*.

We would like to stress the links between the invention of AMSs and the emergence of writing. As mentioned, the main difference between proper writing systems and Paleolithic AMSs lies in the strong linkage with phonology and articulate language of the former. However, writing is, after all, an AMS. One of the main aims of writing at present is the storage and transmission of information; and this almost certainly was its original aim when it was first invented. Paleolithic AMSs fulfilled this function thanks to marks likely referring to specific items relevant to the groups producing them. As we have seen, AMSs may be hypothesized to stem from the cultural exaptation of previous marking tools and techniques, and from the cognitive ability to associate certain engraved objects with some meaning. This may have prompted a further cultural exaptation enabling human beings to associate marks on physical objects not directly to "meanings" but to *spoken words* and, progressively, to elements of the *phonology* of spoken language. This may have enormously increased the amount of information possibly stored, as writing may fully resort to an already existing code: the spoken-language *lexicon*. This scenario may therefore suggest that the emergence of writing is the result of a further step of cultural exaptation joining the abilities characteristic of AMSs with those essential to spoken language. This hypothesis might find further support in the observation that reading is subserved by a brain system joining the lexico-semantic and the grapho-phonological routes. Additionally, the VWFA, lying in the ventral *visual* stream, likely plays a crucial role in integrating these two routes, as we have seen. Therefore, cultural neural reuse of the VWFA for reading might be interpreted both as the outcome of a process of cultural exaptation spanning from AMSs to proper writing systems, and as the way to stabilize such new and complex cultural innovation.

All this sheds new light on the general idea that cultural innovations do not need concomitant genetic evolution as a necessary condition. Indeed, if reading is a key cultural innovation which the human brain may accommodate without the need to evolve, it is reasonable to think that this has also happened for other previous cultural innovations (*ex esse, posse*). One of the main reasons to link astounding cultural innovations to genetic evolution was the more or less implicit assumption that human cognitive endowment heavily depends on our species' neural constitution, which in turn almost deterministically depends on the genetic-developmental program controlling the formation of brain functional anatomy and structural connectivity. Increasing evidence suggests that this "chain of dependence" is much looser, widely plastic, and significantly affected by environmental, social, and cultural factors (Colagè and d'Errico 2018).

Among these factors, teaching likely plays a crucial role, at least for practices like complex tool making, symbolic expressions, AMSs, or writing. Indeed, teaching has recently been emphasized as a key to understanding human cultural evolution (Morgan et al. 2015), and the interrelation between technological behavior, neurocognitive substrates, and the archaeological record (Stout and Khreisheh 2015). It has been convincingly argued that a clear correlation exists between the complexity of the abilities implied by the archaeological record, and the complexity of the required transmission strategies (Högberg et al. 2015). Accordingly, explicit teaching should not be confused with learning by emulation and imitation, or with social learning more generally (Boyd et al. 2011). Teaching, indeed, always involves some form of explanation complementing the learning process. A conceptual framework has recently been proposed (d'Errico and Banks 2015) to classify different transmission strategies of cultural practices used by human and nonhuman primates. The idea behind this framework is not just that cumulative culture implies teaching, but that different transmission strategies exist, many shared with other species, and that the complexification of transmission strategies, up to teaching as we know it nowadays, proceeds in parallel with the complexification of cultural practices and the information that needs to be passed to ensure their transmission. In other words, the emergence of teaching should not be regarded as an all-or-nothing process necessarily triggered by genetic evolution in the human lineage; rather, transmission strategies and teaching evolved gradually and according to *cultural* dynamics influenced by several environmental, social, and accidental factors.

Teaching plays a fundamental role in boosting cultural evolution. According to our model, cultural exaptation sustains the emergence of cultural innovations. This process often leads to diversification and complexification of cultural practices. This in turn requires new transmission strategies to preserve the novelty and ensure its faithful transgenerational transmission. Cultural neural reuse is a mechanism allowing cultural novelties to be acquired, perpetuated, and consolidated at the individual and social level. Cultural neural reuse requires enduring dedicated training to the extent to which, as we have said, it implies the formation of new brain networks and not just the refinement of existing ones. This is evident in the case of reading acquisition, but it should be noted that there is additional preliminary evidence that acquiring other cultural skills through enduring dedicated training induces structural brain changes. For example, teaching present-day humans to produce Paleolithic stone tools, such as Oldowan flakes or Acheulan hand-axes, induces white-matter structural changes in frontoparietal regions (Hecht et al. 2014). Intense training of *adult* macaque monkeys in tool use has been shown to cause formation of new cortico-cortical axonal connections in monkey intraparietal sulcus (Ishibashi et al. 2002; Hihara et al. 2006). In this case, human experimenters imparted the training, as macaque monkeys are neither known to use tools in the wild nor to teach each other. Therefore, appropriate teaching strategies might be essential to trigger cultural neural reuse. We do not know, at present, which cultural transmission strategies may trigger cultural neural reuse. Studies on reading discussed above, and these studies on learning how to make and use tools, suggest that intensive and enduring dedicated training is required to induce this phenomenon. This does not exclude the possibility that other forms of cultural transmission may induce—perhaps to a lesser degree—similar outcomes.

Based on the above, we can predict how the proposed mechanism—encompassing cultural exaptation and cultural neural reuse—should be expected to act on cultural evolution. A cultural innovation would merge two or more existing cultural traits. Adapted modes of cultural transmission must already be in place in the group concerned to allow the new cultural trait to be shared. The cultural innovation should have to do with cultural practices or needs relevant for at least a portion of the population in which it emerges. It does not need to be desirable for *all* individuals, nor to necessarily have a direct survival value; it could just concern a practice in which a part of the population engages frequently for whatever reason. The novel cultural practice should mobilize at least two distinguishable brain networks, previously dedicated to other tasks or practices (those combined at the functional/behavioral level by the novel practice itself).

The prerequisite for this to happen is that the brain of the species in which our mechanism can act should be highly plastic at the ontogenetic level, also beyond the early developmental stages. The specific brain regions involved should also be plastic and adaptable. As far as the example of reading is concerned, it is worth noting that:

- The portion of the fusiform gyrus hosting the VWFA displays an antero-posterior gradient-like organization of sub-areas more and more specialized to recognize frequent letter combinations and words in the individual's language (Vinckier et al. 2007).

- The fusiform gyrus (probably a hominoid-specific brain structure) shows cortical tissue proliferation during face-selective specialization (Gomez et al. 2017). This finding hints at a new model by which emergent brain function and behavior result from cortical tissue proliferation rather than from synaptic selection and strengthening exclusively. This might be crucial for cultural neural reuse (as involving the formation of new anatomical networks) to occur when novel cultural practices are implemented, and is particularly relevant for reading since, as we have seen, the VWFA overlaps the FFA in the left hemisphere.

Brain regions with such characteristics are more likely to support cultural neural reuse.

Finally, it should be emphasized that the mechanism that we propose does not dismiss the role of genetic evolution in cultural evolution. The basic information-processing capabilities of culturally reused regions are outcomes of evolutionary processes. The VWFA exploits brain circuitry previously evolved for object and face recognition, and the fusiform region hosting the VWFA in readers displays high connectivity with other left-hemisphere language areas even prior to reading acquisition (Hannagan et al. 2015; Saygin et al. 2016), and even independently of visual experience (Reich et al. 2011). Patently, these features of the VWFA (and, more generally, of the exalted developmental and ontogenetic plasticity of the human brain) are outcomes of prior biological (and genetic) evolution. However, our hypothesized mechanism substantially differs from processes of gene-culture coevolution. The latter can be regarded, in our lineage, as an extreme case of niche construction heavily conditioned by culture but essentially obeying "classical" dynamics of population genetics (Laland et al. 1999, 2010; Odling-Smee et al. 2003). We emphasize that cultural dynamics played a driving role upon the suite of cultural adaptations associated with the evolution of our lineage (Jablonka and Lamb 2005; Dor and Jablonka 2014). We put more emphasis on genuine cultural processes, and argue that the emergence of the cultural innovations characterizing our species is better explained as the result of cultural dynamics acting at a level that does not involve the production of *heritable* genetic variability and *selection* of successful variants.

In other words, the proposed mechanism based on cultural exaptation, teaching, and cultural neural reuse may ensure the stabilization and spreading of cultural innovations without the need of concomitant genetic evolution. It is theoretically not impossible that on longer timescales, the innovations stabilized through the proposed mechanism become somehow "genetically assimilated" (Waddington 1942, 1953; see also Varki et al. 2008), and this may well happen via gene-culture coevolution. However, this would be a *subsequent* stage that is not strictly required for significant cultural innovations to become stabilized, spread, and transmitted from generation to generation.

Conclusion

We argue that repeated cycles of cultural exaptation, development of appropriate strategies of cultural transmission, and ensuing cultural neural reuse represent the fundamental mechanism that has regulated the cultural evolution of our lineage. This mechanism accounts for the asynchronous and patchy emergence of innovations around the globe. It also serves as a satisfactory explanation for trends toward complexification of cultural practices recorded at different times and places in human history. It is consistent with experimental results that have demonstrated the impact of intense training and teaching on brain anatomy and connectivity when learning skills widespread in Paleolithic times (e.g., knapping) or more recently (e.g., reading).

Future research should explore ways to further test the pertinence of the proposed mechanism, and design strategies to figure out the role played by each of its three components—cultural exaptation, cultural transmission strategies, cultural neural reuse—and their interaction. Inquiry into cultural exaptation may use historical, ethnographic, and archaeological instances to document in detail the processes leading to shifts in function and creation of new cultural practices. Similar to processes of microevolution within a species, cultural changes leading to innovations in the technological and symbolic domains can occur via a punctuated process that may have left detectable traces in the ethnographic and archaeological record. By documenting those intermediate steps and identifying the ones leading to a significant shift in function we may explain the way in which exaptation takes place and novel cultural practices spread in a community. Ethnographic data, experimental archaeology, and technological studies may unveil the behavioral patterns stemming from episodes of cultural exaptation, and allow us to predict the necessary minimal teaching requirements (d'Errico and Banks 2015).

Current knowledge of brain functional anatomy can help with predicting the regions and networks involved in neural reuse, facilitating the stabilization and transmission of specific novel cultural practices. However, dedicated experiments, such as those assessing the consequences of knapping-skill learning on brain structure, will be essential to precisely establish how new learning practices affected neural organization during the cultural evolution of our lineage.

Application of such an integrated research strategy may be instrumental to understand how the components of the proposed mechanism interacted in the past, establish whether significant changes in these interactions occurred at some point, and possibly identify thresholds that have allowed the definite establishment of cumulative culture as we know it.

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